AS · Unit 1 · WBI11 Molecules, Diet, Transport and Health

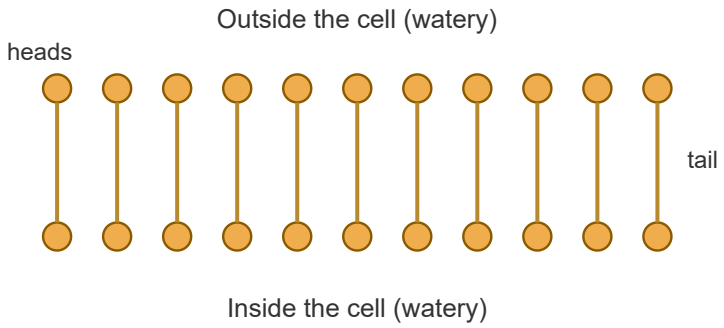
Membranes and transport

Every cell is enclosed by a membrane that decides what gets in and what gets out. This topic explains how the membrane is built from phospholipids and proteins, and how substances cross it by passive movement, by using energy, or by being packed into vesicles.

1. Why membranes form a bilayer

A phospholipid has two very different ends. The head is a phosphate group that is charged and attracted to water, so it is hydrophilic. The two tails are long fatty acid chains with no charge, so they are hydrophobic and avoid water. When phospholipids are placed in water they arrange themselves so that the heads face the watery solution on each side and the tails point inwards, away from it. The result is a double layer, or bilayer, with a water-hating core.

This arrangement is the reason membranes work as barriers. The hydrophobic core stops most charged ions and polar molecules from simply dissolving through the membrane, while small non-polar molecules can slip between the tails.



- **Phospholipid:** a molecule with a polar, hydrophilic phosphate head and two non-polar, hydrophobic fatty acid tails.
- **Phospholipid bilayer:** the double layer of phospholipids that forms the basic framework of every cell membrane.

2. The fluid mosaic model

Early models were simpler. The Gorter and Grendel model proposed a single phospholipid bilayer, but it could not explain how a membrane holds together its whole cell contents. The Davson-Danielli sandwich model placed a layer of protein on each face of a phospholipid core. This explained the membrane's strength but implied that all membranes look alike and that the proteins sit only on the surface.

The fluid mosaic model of Singer and Nicolson replaced these ideas. The membrane is a fluid sheet of phospholipids in which proteins are scattered like tiles in a mosaic. Phospholipids and many proteins can move sideways within their own layer, which is why the membrane is described as fluid. The model was supported by freeze-fracture images, which split the membrane down its middle and showed protein particles set into a smooth

matrix. It was also supported by experiments in which the membrane proteins of a mouse cell and a human cell mixed over time in a hybrid cell.

KEY IDEA

The Davson-Danielli model has proteins only on the surface and suggests a uniform structure. The fluid mosaic model has proteins that are embedded within the bilayer, and it allows the membrane to be fluid and varied. Both models have a phospholipid bilayer, so the difference lies in where the proteins are and how they move.

3. The parts of the membrane and what they do

Membrane proteins come in two broad groups. Integral proteins are embedded in the hydrophobic core of the bilayer, and many of them span the whole membrane. Peripheral proteins are not inside the bilayer at all. They are held loosely on its surface, often attached to the integral proteins. Among the integral proteins, channel proteins form water-filled pores that let polar molecules and ions through, while carrier proteins bind a specific molecule and change shape to move it across.

Cholesterol sits between the phospholipid tails. At warm temperatures it stops the phospholipids from moving so freely, which makes the membrane less fluid. At cool temperatures it stops the tails from packing tightly, which keeps the membrane fluid. Cholesterol is therefore a buffer that keeps the membrane working over a range of temperatures, and it also reduces the leakage of small polar molecules.

Carbohydrates are attached on the outer surface of the membrane. Where a carbohydrate is bound to a lipid this forms a glycolipid, and where it is bound to a protein it forms a glycoprotein. Together they make up the glycocalyx. These carbohydrate chains act as recognition markers, letting cells identify one another and receptor sites detect chemical signals.

Component	What it is	Main role
Phospholipid bilayer	Two layers of phospholipids	Forms a selectively permeable barrier
Integral protein	Embedded in the bilayer, often spanning it	Channels and carriers for transport
Peripheral protein	Loosely attached to the surface	Support and signalling
Cholesterol	Sits between the tails	Keeps fluidity steady at different temperatures
Glycolipid and glycoprotein	Carbohydrate bound to lipid or protein	Cell recognition and receptor sites

COMMON MISTAKE

Do not describe the proteins in the membrane as floating freely in a sea of lipid with no structure. The bilayer is organised: heads face water, tails face inwards, and the proteins are arranged around that core.

4. Passive transport: diffusion

Diffusion is the net movement of particles from a region of higher concentration to a region of lower concentration, down a concentration gradient. It happens because particles move randomly, so there are more movements away from a crowded area than into it. It needs no energy from the cell, which is why it is called passive.

Small non-polar molecules such as oxygen and carbon dioxide, and hydrophobic molecules such as benzene, pass straight through the phospholipid bilayer. Charged ions and large polar molecules cannot, because the hydrophobic core repels them. The rate of diffusion depends on the temperature (faster when hotter, because particles have more kinetic energy), the size of the concentration gradient, the surface area available, the size of the particles (smaller particles move faster), and the medium (diffusion is fastest in gases and slowest in solids).

Fick's law combines three of these factors for an exchange surface such as the lining of the alveoli: rate of diffusion $\propto$ (surface area $\times$ difference in concentration) $\div$ thickness of the surface.

The surface available for exchange depends on an organism's size, as the surface area to volume ratio of a cube shows.

5. Passive transport: facilitated diffusion

Some substances are polar or charged, so they cannot cross the bilayer on their own, but they still move down their gradient. Facilitated diffusion uses a membrane protein to do this. Channel proteins open a water-filled route for ions or polar molecules. Carrier proteins bind the molecule on one side, change shape, and release it on the other side. Facilitated diffusion is still passive: it goes from high to low concentration and needs no ATP. It is faster than simple diffusion for the molecules it serves, but it is limited by the number of proteins available.

The cell can speed up or slow down the transport by changing how many carrier or channel proteins it places in the membrane. An example is glucose leaving liver cells into the blood through carrier proteins, which takes place without any energy input.

6. Osmosis and water potential

Osmosis is the net movement of water molecules across a partially permeable membrane. Water moves from a region where its concentration is higher to a region where it is lower. A dilute solution has a high concentration of free water, and a concentrated solution has a low one, because the dissolved solute takes up space and attracts some of the water. Water therefore moves from the dilute side to the concentrated side. Its movement can be described in terms of water potential, which is higher in dilute solutions and lower in concentrated ones.

What happens to a cell depends on the solution around it. In an isotonic solution the water movement in and out is balanced and the cell keeps its shape. In a hypotonic solution water enters: an animal cell may swell and burst, while a plant cell becomes turgid because the cell wall stops it from bursting. In a hypertonic solution water leaves: an animal cell shrinks, and a plant cell's cytoplasm pulls away from the wall, which is called plasmolysis.

Predicting what happens to a plant cell in a solution

1. Identify the water potential of the solution compared with the cell. A cell's contents usually have a lower water potential than pure water.
2. If the solution is more dilute than the cell (hypotonic), water moves in by osmosis, the vacuole fills, and the cell becomes turgid against the wall.
3. If the solution is more concentrated than the cell (hypertonic), water moves out, the vacuole shrinks, and the cytoplasm pulls away from the wall. This is plasmolysis.
4. Explain each step by pointing to the direction of the water potential gradient across the partially permeable membrane.

7. Active transport

Active transport moves molecules against their concentration gradient, from low concentration to high concentration. This is uphill, so the cell must supply energy. The energy usually comes from ATP. The substance to be moved binds to a carrier protein, ATP is used to change the shape of the protein, and the protein releases the substance on the other side of the membrane. Because the process is directed, it moves in one direction only. Active transport is needed when the cell must keep a high concentration of a substance inside, such as potassium ions in cells, or must remove a substance that is at a lower concentration outside, such as absorbing mineral ions from the soil into root hair cells.

KEY IDEA

Active transport always needs energy and always uses carrier proteins. Facilitated diffusion uses protein but no energy, and always runs down the gradient.

Feature	Facilitated diffusion	Active transport
Direction	Down the concentration gradient	Against the concentration gradient
Energy needed	No	Yes (ATP or an ion gradient)
Membrane proteins	Channel or carrier proteins	Carrier proteins
Example	Glucose leaving liver cells	Mineral ions entering root hair cells

8. Bulk transport: endocytosis and exocytosis

Very large molecules, or many molecules at once, are moved across the membrane in vesicles. In endocytosis the membrane folds inwards around material outside the cell, and a new vesicle pinches off inside the cytoplasm. In exocytosis a vesicle, often one made by the Golgi apparatus, moves to the membrane and fuses with it, releasing its contents outside. Both processes use energy from ATP and depend on the membrane's fluidity, which lets it bend and fuse.

EXAM TIP

When you name a process in an exam, match the direction to the name: endocytosis brings material in, exocytosis sends it out.

9. Summary

Process	Direction	Energy	Protein needed	Example
Diffusion	High to low concentration	No	No	Oxygen moving into cells
Facilitated diffusion	High to low concentration	No	Yes	Glucose entering cells
Osmosis	Water from dilute to concentrated	No	Water passes through the bilayer or channels	Water entering kidney tubules
Active transport	Low to high concentration	Yes (ATP)	Yes	Mineral ions into root hair cells
Endocytosis and exocytosis	Into or out of the cell	Yes (ATP)	Vesicle formation	Secretion from secretory cells

10. Quick checks

QUICK CHECK

Which molecule is most likely to cross a phospholipid bilayer by simple diffusion?

- Sodium ions
- Glucose
- Oxygen
- Amino acids

Oxygen is small and non-polar, so it can dissolve in the hydrophobic core of the bilayer. Sodium ions are charged and glucose is a large polar molecule, so both need a protein. Amino acids are polar and cannot pass the core unaided.

QUICK CHECK

A red blood cell is placed in a very dilute salt solution. What is the most likely result?

- It shrinks as water leaves
- It swells and may burst as water enters
- It stays the same size because the solution is isotonic
- It becomes turgid because it has a cell wall

A dilute solution is hypotonic to the cell, so water enters by osmosis. Animal cells have no wall to resist the increase in volume, so they swell and can burst. The turgid answer applies to plant cells, which have a wall.

QUICK CHECK

Which statement best describes active transport?

- Movement down a gradient using a channel protein
- Movement of water through a partially permeable membrane
- Movement against a gradient, using carrier proteins and energy from ATP
- Bulk movement of solid particles through a single pore

Active transport moves substances against the gradient, so it needs ATP and carrier proteins that change shape. Channel-based movement down the gradient is facilitated diffusion, and osmosis concerns water only.

QUICK CHECK

Why does cholesterol help keep a membrane working across a range of temperatures?

- It makes the bilayer fully solid at low temperatures
- It pumps ions across the membrane
- It restrains phospholipid movement when warm and prevents tight packing when cool
- It attaches carbohydrates to the outer surface

Cholesterol stops the membrane becoming too fluid at high temperatures and too rigid at low temperatures. It does not make the membrane solid, pump anything, or attach carbohydrates; those roles belong to other molecules.